USN | | | | | | | | | | [22EC32]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER - 2023

ANALOG AND DIGITAL CIRCUITS - 2

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. **ANSWER ALL QUESTIONS**

		PART – A			**5 X 2=10**
1.	**a.**	Define CMRR of op amp and specify the ideal value.	L2	CO1	**02**
	b.	Write the internal block diagram of the op amp.	L2	CO1	**02**
	c.	Construct block diagram with frequency response to realize the notch filter using band pass filter.	L3	CO2	**02**
	d.	What is the analog voltage equivalent of binary input {1 1 1} of 3 bit DAC for reference voltage of 10v.	L3	CO4	**02**
	e.	A memory that has a storage capacity of 4K * 20, How many words can be stored on this chip? What is the word size? How many total bits can this chip store?	L3	CO5	**02**
		PART - B			**6 X 5=30**
2.	**a.**	With a neat circuit diagram, derive an expression for gain of non-inverting amplifier.	L3	CO1	**05**
		OR			
	b.	Discuss the ideal characteristics of the op amp with op amp model.	L2	CO1	**05**
3.	**a.**	Discuss different types of filters and plot the ideal and practical frequency responses.	L2	CO2	**05**
		OR			
	b.	Discuss RC phase shift oscillator by constructing the circuit diagram and highlight the condition for oscillations.	L3	CO2	**05**
4.	**a.**	Discuss with a block diagram the system showing the application of ADC and DAC converters.	L2	CO4	**05**
		OR			
	b.	Discuss with a neat diagram, Schmitt trigger operation and waveform generation using 555 timer.	L3	CO4	**05**
5.	**a.**	With a block diagram of 16x8 ROM, describe the inputs, control inputs and output lines.	L3	CO5	**05**

		OR			
	b.	Discuss the features of PROM with transistor diagram of fusible links.	L3	CO5	**05**
6.	**a.**	i. Design active second order low pass filter for a cutoff frequency of 5KHz and gain of 2. ii. Plot the frequency response and highlight the cut off frequency and roll off rate.	L3	CO6	**05**
		OR			
	b.	i. Design active second order high pass filter for a cutoff frequency of 8KHz and gain of 3.	L6	CO6	**05**
		ii. Plot the frequency response and highlight the cut off frequency and roll off rate.			
7.	**a.**	Discuss the operation of op amp based differentiator with circuit diagram and waveforms.	L4	CO2	**05**
		OR			
	b.	Discuss the operation of op amp based integrator with circuit diagram and waveforms.	L4	CO2	**05**
		PART - C		**6 X 10=60**	
8.	**a.**	With a schematic diagram of successive approximation ADC, explain the principle of operation with suitable example.	L4	CO4	**07**
	b.	Construct the flow chart to realize the operation of successive approximation ADC.	L3	CO4	**03**
		OR			
	a.	Discuss the operation of op amp based sample and hold circuit with waveforms.	L4	CO2	**06**
	b.	Explain the realization of voltage to current converter circuit using op amp.	L3	CO2	**04**
9.	**a.**	Construct and discuss the functionality of the square wave generator circuit.	L4	CO2	**10**
		OR			
	b.	Construct the triangular wave generator circuit and also discuss it's operation.	L4	CO2	**10**

10.	**a.**	Explain the following ADC/DAC specifications **i.** Linearity **ii.** Accuracy **iii.** Resolution **iv.** Monotonocity **v.** Settling time	L3	CO4	**10**
		OR			
	b.	Explain the principle of operation of weighted resistor type DAC with a circuit diagram.	L4	CO4	**10**
11.	**a.**	Discuss the features of RAM with a block diagram of internal organization of 64X4 memory.	L4	CO5	**10**
		OR			
	b.	Discuss the following terms with respect to logic families **i.** Logic levels **ii.** Propagation delay **iii.** Power dissipation **iv.** Fan in and fan out **v.** Noise margin	L2	CO5	**10**
12.	**a.**	Design a monostable multivibrator for a time period of 10msec using 555 timer with a neat functional diagram.	L6	CO6	**10**
		OR			
	b.	Design astable multivibrator using 555 timer for T=10msec and 50% duty cycle with a neat functional diagram.	L6	CO6	**10**
13.	**a.**	Construct and explain the principle of operation of multiplier and divider circuit using op amp.	L4	CO4	**10**
		OR			
	a.	Discuss the features of TTL and CMOS logic families.	L3	CO5	**05**
	b.	Discuss the features of PAL, PLA and FPGA.	L3	CO5	**05**